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Psymple: A Python package for complex systems modelling

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Summary

The modelling of complex systems, which are characterised by high degrees of interdependence amongst components, is continually evolving in response to developing computing and visualisation power. Examples such as ecological, economic or social systems are classically modelled using statistical or correlative techniques, which have natural shortcomings when used for predictive modelling outside of the range of their parameterisation data.

Predictive modelling is improved by deterministic models which capture the evolution of a system, or by hybrid models composed of both deterministic and correlative parts. These models often feature significantly increased size and complexity, and there has been a growing requirement for so-called next generation modelling platforms which are built to a specification requiring modular and data-first implementations which drive clear, adaptable, reusable and accessible modelling practices.

The platform `psymple` is designed to facilitate the development of hybrid complex system models and modelling frameworks. It allows users to link together arbitrary combinations of modular differential equation systems and functional components to build categorical diagrams representing a complex system. A compilation process automatically generates simulatable system equations from these diagrams using the symbolic mathematics package `sympy`. Ultimately, this allows users to focus on the components and interactions of models, rather than their complex equation structure.

Statement of need

The development of `psymple` emerged from the complex system modelling requirements of ecological systems. Ecological niche models, also called species distribution models (SDMs) predict species distributions in response to environmental variables such as geographic and climatic features (Elith & Franklin, 2017; Elith & Leathwick, 2009). Classically, these models are formed using correlative or statistical approaches which match observational data to a set of environmental variables to produce favourability ranges for each species.

An alternative approach is mechanistic modelling, which instead of observational data use physiological data to capture the underlying mechanisms which drive species distribution, such as energy balance, population dynamics or response to climate (Kearney & Porter, 2009). In contrast to correlative approaches, mechanistic SDMs decouple the physiology of a species from their native geography or climate, and allow SDMs in new geographic or climatic regimes to be created in the absence of observational data (Johnston et al., 2019).

An example mechanistic framework is physiologically-based demographic modelling (PBDM), which creates holistic ecosystem models based on the weather-driven biology of component species, allowing for predictive phenology, age- or mass-structured population dynamics, and distribution assessments, see (Gutierrez & Ponti, 2022a, 2022b) for an overview and further references. With this approach, PBDM can account for tritrophic ecosystem interactions (Gutierrez et al., 1999), or model the effects of climate change (Gutierrez et al., 2023).

The schools of thought around correlative SDMs and mechanistic frameworks such as PBDM are seen as largely disjoint (Dormann et al., 2012), but their development can be traced back to early common roots (Fitzpatrick & Nix, 1970; Gutierrez et al., 1974; Wit & Goudrian, 1978). A component of the PBDM framework is the use of physiological data to parametrise "biodemographic" functions capturing biophysical or biochemical mechanisms, such as the development, mortality and fecundity rates of a species in response to environmental variables (Ponti & Gutierrez, 2023).

The use of biodemographic functions in PBDM combines the considerable holistic advantages of mechanistic SDMs, while retaining the comparable ease of parametrisation as status-quo correlative models. More widely, there has been growing interest and development of ecological models explicitly composed of both correlative and mechanistic components (Buckley et al., 2010; Tourinho & Vale, 2023), combining the benefits of both areas. In the wider context of complex systems modelling, the approach of building composite models out of different techniques is called hybrid, or spectrum, modelling.

While PBDM is a highly-developed framework, it shares the same barriers to widespread introduction as general hybrid complex system models. These barriers include the lack of available modelling frameworks (Buckley et al., 2018), the lack of flexibility in existing models (Buckley et al., 2010), or the lack of modelling platforms to implement existing ideas (Ponti & Gutierrez, 2023). The package `psymple` is a general platform designed to facilitate the creation of hybrid complex systems models and modelling frameworks.

Models in `psymple` are built from arbitrary combinations of modular mechanistic, dynamic components and correlative, functional components which naturally interact with each other. This structure allows for the systematic implementation of modelling frameworks such as PBDM, and, more widely, those capturing biological, economic and social systems, for which it is not feasible to capture the laws of interaction purely mechanistically. Examples include agroecological, bioeconomic and Earth systems modelling, and the development of `psymple` is a necessary first step in the development and release of accessible and impactful tools in these areas.

Description and related software

The workings of `psymple` are based on the ideas of the dynamic systems modelling package `AlgebraicJulia.jl` (Libkind et al., 2024), in which modular objects, called *resource sharers*, containing ordinary differential equations, are linked by wires which represent the sharing of state variables across the dynamics equations of the system. A compilation process based in mathematical category theory (Libkind et al., 2022) maps this diagram of objects and wires into a complex interacting dynamical system.

In `psymple`, these ideas are extended to allow for a second type of object which computes multivariate functions. Wires between these objects formally represent partial functional substitution, and together with dynamic resource sharers enable arbitrary hybrid models of complex systems to be created, compiled, and simulated. Underneath, the mathematical conversion from formal system diagrams of differential equations and functions to simulatable systems is handled by the Python symbolic mathematics package `sympy` (Meurer et al., 2017).

Additionally, `psymple` enables arbitrary nesting of these base objects inside composite objects whose hierarchy represents modelled structure. With these structures, `psymple` satisfies

a specification shared by "next-generation" dynamical systems modelling frameworks, see (Baez et al., 2023), including being *faceted*, where models can be considered one piece at a time; *modular*, where components naturally compose together; and *functorial*, where the data describing the model (its syntax) is systematically and reliably transformed into system behaviour (its semantics). These ideas allow for legible modelling of highly complex, specialised systems, and drive clear, adaptable and accessible modelling practices in line with those promoted by the Open Modeling Foundation.

Many software solutions exist to capture climate-driven behaviour within parts of agricultural ecosystems. For example, sophisticated climate-sensitive crop models can be produced using `APSIM` (Holzworth et al., 2014) or `DSSAT` (Jones et al., 2003). For insects, `ILCYM` (Sporleder et al., 2009) enables the creation of climate-driven pest insect phenology models for distribution and risk assessments. The goal of `psymple` and its framework is not to stand apart from this existing software, but instead enable the creation of software which can link between existing tools and enable such assessments across ecosystems, including the plant, herbivore and natural enemy trophic levels and their interactions.

Related research

The development of `psymple` emerged as part of a collaborative effort between IDEMS International and CASAS Global to increase the uptake, accessibility and impact of the physiologically-based demographic modelling (PBDM) framework. The development of `psymple` is a necessary first step to capture PBDM in sufficient generality to enable modular and flexible implementations of existing and new models into the framework to drive future research and impact potential.

The specification of `psymple` was created from the "collaborative modelling" approach headed-up by the Topos Institute, which uses the mathematical field of applied category theory to capture the mechanisms required to create modelling frameworks collaboratively, accounting for diversity in technical language, data availability and subject expertise. This specification, coupled with the generality and flexibility of `psymple`, enables frameworks beyond PBDM to be developed or implemented, and pushed towards new research and impacts.

For example, research is currently active in bioeconomic modelling, which extends the supply-demand and mass-structured mechanisms of PBDM to understand the economic interaction that humans have with agroecological systems, see (Gutierrez et al., 2020; Regev et al., 1998). More widely, ideas are being conceived around applications in Earth systems modelling, multi-layer modelling incorporating feedback loops to modelling techniques such as collaborative agent-based modelling, and data management and source control across collaborative practices.

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References

Baez, J., Li, X., Libkind, S., Osgood, N. D., & Patterson, E. (2023). Compositional modeling with stock and flow diagrams. *Electronic Proceedings in Theoretical Computer Science*, *380*, 77–96. https://doi.org/10.4204/eptcs.380.5

Buckley, L., Cannistra, A., & John, A. (2018). Leveraging Organismal Biology to Forecast the Effects of Climate Change. *Integrative and Comparative Biology*, *58*(1), 38–51. https://doi.org/10.1093/icb/icy018

Buckley, L., Urban, M., Angilletta, M., Crozier, L., Rissler, L., & Sears, M. (2010). Can mechanism inform species' distribution models? *Ecology Letters*, *13*(8), 1041–1054. https://doi.org/10.1111/j.1461-0248.2010.01479.x

Dormann, C. F., Schymanski, S. J., Cabral, J., Chuine, I., Graham, C., Hartig, F., Kearney, M., Morin, X., Römermann, C., Schröder, B., & Singer, A. (2012). Correlation and process in species distribution models: Bridging a dichotomy. *Journal of Biogeography*, *39*(12), 2119–2131. https://doi.org/10.1111/j.1365-2699.2011.02659.x

Elith, J., & Franklin, J. (2017). *Species distribution modeling*. https://doi.org/10.1016/B978-0-12-809633-8.02390-6

Elith, J., & Leathwick, J. R. (2009). Species distribution models: Ecological explanation and prediction across space and time [Journal Article]. *Annual Review of Ecology, Evolution, and Systematics*, *40*(Volume 40, 2009), 677–697. https://doi.org/10.1146/annurev.ecolsys.110308.120159

Fitzpatrick, E., & Nix, H. (1970). *The climatic factor in Australian grassland ecology*. 3–26. https://api.semanticscholar.org/CorpusID:135291467

Gutierrez, A. P., Nix, H., Havenstein, D., & Moore, P. (1974). The ecology of *Aphis craccivora* koch and subterranean clover stunt virus in south-east australia. III. A regional perspective of the phenology and migration of the cowpea aphid. *Journal of Applied Ecology*, *11*(1), 21–35. http://www.jstor.org/stable/2402002

Gutierrez, A. P., & Ponti, L. (2022a). Analysis of invasive insects: Links to climate change. In *Invasive Species and Global Climate Change* (Second, pp. 50–73). https://doi.org/10.1079/9781800621459.0000

Gutierrez, A. P., & Ponti, L. (2022b). Assessing and managing the impact of climate change on an invasive weed, yellow starthistle. In *Invasive Species and Global Climate Change* (Second, pp. 260–281). https://doi.org/10.1079/9781800621459.0000

Gutierrez, A. P., Ponti, L., Kranthi, K., Baumgärtner, J., Kenmore, P., Gilioli, G., Boggia, A., Cure, J., & Rodríguez, D. (2020). Bio-economics of indian hybrid bt cotton and farmer suicides. *Environmental Sciences Europe*, *32*, 139. https://doi.org/10.1186/s12302-020-00406-6

Gutierrez, A. P., Sabbatini Peverieri, G., Ponti, L., Giovannini, L., Roversi, P., Mele, A., Pozzebon, A., Scaccini, D., & Hoelmer, K. (2023). Tritrophic analysis of the prospective biological control of brown marmorated stink bug, *Halyomorpha halys*, under extant weather and climate change. *Journal of Pest Science*, *96*, 1–22. https://doi.org/10.1007/s10340-023-01610-y

Gutierrez, A. P., Yaninek, J., Neuenschwander, P., & Ellis, C. K. (1999). A physiologically-based tritrophic metapopulation model of the african cassava food web. *Ecological Modelling*, *123*(2), 225–242. https://doi.org/10.1016/S0304-3800(99)00144-1

Holzworth, D. P., Huth, N. I., deVoil, P. G., Zurcher, E. J., Herrmann, N. I., McLean, G., Chenu, K., & others. (2014). APSIM – evolution towards a new generation of agricultural systems simulation. *Environmental Modelling & Software*, *62*, 327–350. https://doi.org/10.1016/j.envsoft.2014.07.009

Johnston, A. S. A., Boyd, R. J., Watson, J. W., Paul, A., Evans, L. C., Gardner, E. L., & Boult, V. L. (2019). Predicting population responses to environmental change from individual-level mechanisms: Towards a standardized mechanistic approach. *Proceedings of the Royal Society B: Biological Sciences*, *286*(1913), 20191916. https://doi.org/10.1098/

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rspb.2019.1916

Jones, J. W., Hoogenboom, G., Porter, C. H., Boote, K. J., Batchelor, W. D., Hunt, L. A., Wilkens, P. W., Singh, U., Gijsman, A. J., & Ritchie, J. T. (2003). The DSSAT cropping system model. *European Journal of Agronomy*, *18*(3-4), 235–265.

Kearney, M., & Porter, W. (2009). Mechanistic niche modelling: Combining physiological and spatial data to predict species' ranges. *Ecology Letters*, *12*(4), 334–350. https://doi.org/10.1111/j.1461-0248.2008.01277.x

Libkind, S., Baas, A., Patterson, E., & Fairbanks, J. (2022). Operadic modeling of dynamical systems: Mathematics and computation. *Electronic Proceedings in Theoretical Computer Science*, *372*, 192–206. https://doi.org/10.4204/EPTCS.372.14

Libkind, S., Baas, A., Patterson, E., & Fairbanks, J. (2024). AlgebraicDynamics.jl. In *GitHub repository*. GitHub. https://github.com/AlgebraicJulia/AlgebraicDynamics.jl

Meurer, A., Smith, C. P., Paprocki, M., Čertík, O., Kirpichev, S. B., Rocklin, M., Kumar, A., Ivanov, S., Moore, J. K., Singh, S., Rathnayake, T., Vig, S., Granger, B. E., Muller, R. P., Bonazzi, F., Gupta, H., Vats, S., Johansson, F., Pedregosa, F., … Scopatz, A. (2017). SymPy: Symbolic computing in python. *PeerJ Computer Science*, *3*, e103. https://doi.org/10.7717/peerj-cs.103

Ponti, L., & Gutierrez, A. P. (2023). Challenging the status quo in invasive species assessment using mechanistic physiologically based demographic modeling. *Environment, Development and Sustainability*. https://doi.org/10.1007/s10668-023-03698-9

Regev, U., Gutierrez, A. P., Schreiber, S. J., & Zilberman, D. (1998). Biological and economic foundations of renewable resource exploitation. *Ecological Economics*, *26*(3), 227–242. https://doi.org/10.1016/S0921-8009(97)00103-1

Sporleder, M., Chavez, D., Gonzales, J., Juarez, H., Simon, R., & Kroschel, J. (2009, November). *ILCYM-insect life cycle modeling: Software for developing temperature-based insect phenology models with applications for regional and global pest risk assessments and mapping*.

Tourinho, L., & Vale, M. M. (2023). Choosing among correlative, mechanistic, and hybrid models of species' niche and distribution. *Integrative Zoology*, *18*(1), 93–109. https://doi.org/10.1111/1749-4877.12618

Wit, C. de, & Goudrian, J. (1978). *Simulation of ecological processes* (2nd ed.). Centre for Agricultural Publishing; Documentation, Wageningen, The Netherlands. https://doi.org/10.1016/0304-3800(80)90048-4

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